



FACULTÉ DES SCIENCES ET DES TECHNOLOGIES(FST)

TD N°_4 – SYSTÈME

COURS: SYSTÈME

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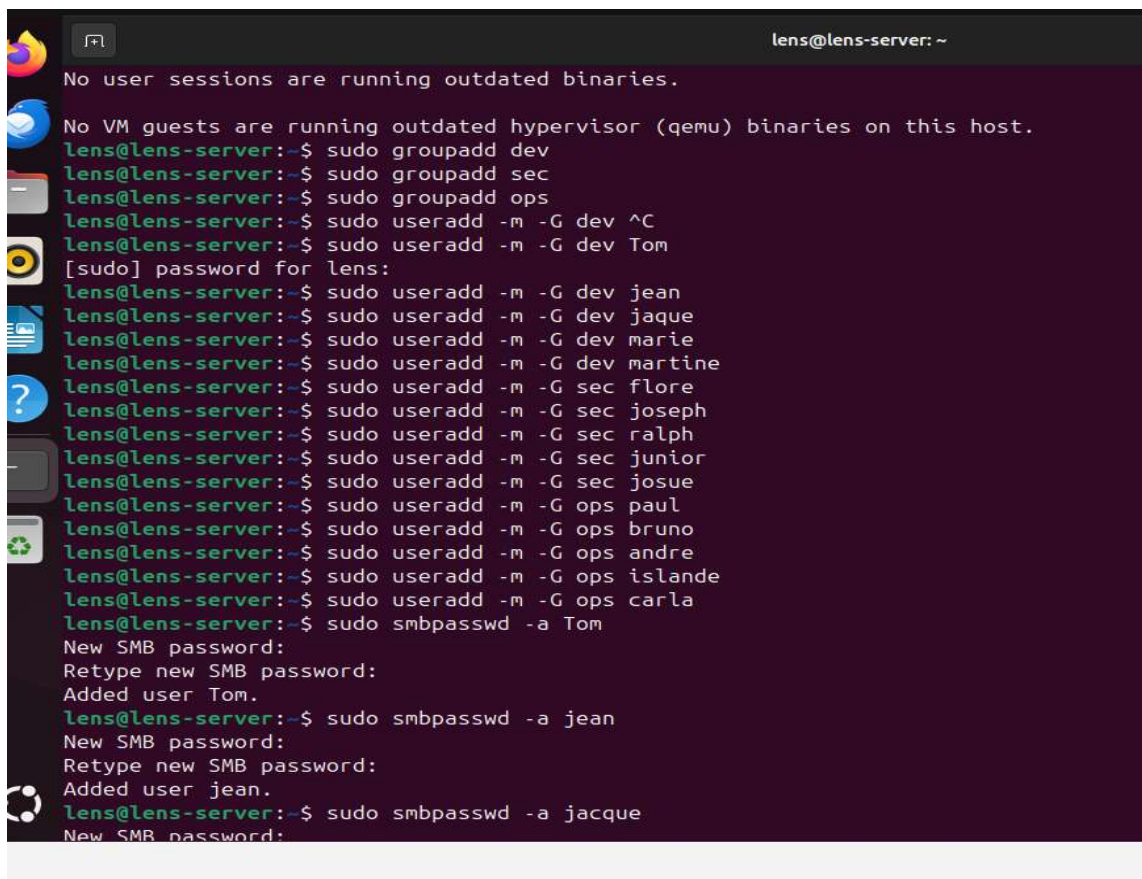
L'objectif de ce TD est de:

1. Comprendre les comptes utilisateurs et groupes sous Linux
2. Créer, modifier, supprimer utilisateurs et groupes
3. Gérer les permissions : r, w, x, chmod, chown, chgrp
4. Configurer les permissions avancées : SUID, SGID, sticky bit
5. Savoir vérifier et auditer les droits

1-Création des groupes.

2- Crédes utilisateurs avec ajout automatique au bon groupe.

Les deux dans une même image.

A terminal window titled 'lens@lens-server: ~' showing a series of commands to create groups and users. The terminal output includes messages about outdated binaries and the successful creation of users 'jean', 'jaque', 'marie', 'martine', 'flore', 'joseph', 'ralph', 'junior', 'josue', 'paul', 'bruno', 'andre', 'islande', and 'carla' under the 'dev' and 'ops' groups, and the user 'Tom' under the 'sec' group. The terminal also shows the creation of SMB passwords for 'Tom', 'jean', and 'jaque'.

```
lens@lens-server: ~  
No user sessions are running outdated binaries.  
No VM guests are running outdated hypervisor (qemu) binaries on this host.  
lens@lens-server:~$ sudo groupadd dev  
lens@lens-server:~$ sudo groupadd sec  
lens@lens-server:~$ sudo groupadd ops  
lens@lens-server:~$ sudo useradd -m -G dev ^C  
lens@lens-server:~$ sudo useradd -m -G dev Tom  
[sudo] password for lens:  
lens@lens-server:~$ sudo useradd -m -G dev jean  
lens@lens-server:~$ sudo useradd -m -G dev jaque  
lens@lens-server:~$ sudo useradd -m -G dev marie  
lens@lens-server:~$ sudo useradd -m -G dev martine  
lens@lens-server:~$ sudo useradd -m -G sec flore  
lens@lens-server:~$ sudo useradd -m -G sec joseph  
lens@lens-server:~$ sudo useradd -m -G sec ralph  
lens@lens-server:~$ sudo useradd -m -G sec junior  
lens@lens-server:~$ sudo useradd -m -G sec josue  
lens@lens-server:~$ sudo useradd -m -G ops paul  
lens@lens-server:~$ sudo useradd -m -G ops bruno  
lens@lens-server:~$ sudo useradd -m -G ops andre  
lens@lens-server:~$ sudo useradd -m -G ops islande  
lens@lens-server:~$ sudo useradd -m -G ops carla  
lens@lens-server:~$ sudo smbpasswd -a Tom  
New SMB password:  
Retype new SMB password:  
Added user Tom.  
lens@lens-server:~$ sudo smbpasswd -a jean  
New SMB password:  
Retype new SMB password:  
Added user jean.  
lens@lens-server:~$ sudo smbpasswd -a jacque  
New SMB password:
```

3. Mise en place de l'arborescence.

```
? 0 directories, 0 files
lens@lens-server:~$ sudo mkdir -p /srv/samba/dev
lens@lens-server:~$ sudo mkdir -p /srv/samba/sec
lens@lens-server:~$ sudo mkdir -p /srv/samba/ops
lens@lens-server:~$ sudo mkdir -p /srv/samba/shared
lens@lens-server:~$ sudo mkdir -p /srv/samba/collab-dev-sec
lens@lens-server:~$ sudo mkdir -p /srv/samba/public
lens@lens-server:~$ sudo tree /srv/samba
/srv/samba
├── collab-dev-sec
├── dev
├── ops
├── public
├── sec
└── shared

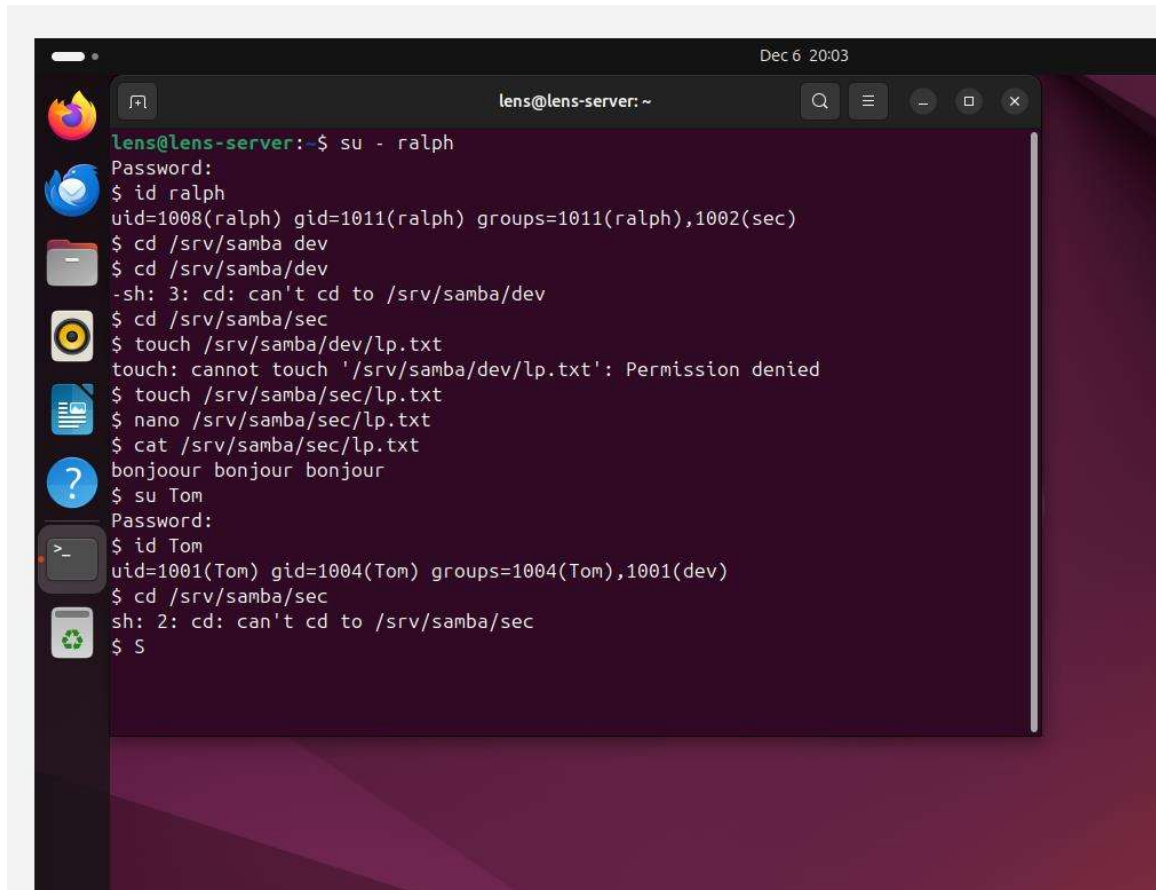
7 directories, 0 files
lens@lens-server:~$
```

4. Configuration des permissions.

```
7 directories, 0 files
lens@lens-server:~$ sudo chmod -R 777 /srv/samba/public
lens@lens-server:~$ sudo chown -R :dev /srv/samba/dev
lens@lens-server:~$ sudo chmod 2770 /srv/samba/dev
lens@lens-server:~$ sudo chown -R :sec /srv/samba/sec
lens@lens-server:~$ sudo chmod 2770 /srv/samba/sec
lens@lens-server:~$ sudo chown -R ;ops /srv/samba/ops
chown: missing operand
Try 'chown --help' for more information.
Command 'ops' not found, but there are 18 similar ones.
lens@lens-server:~$ sudo chown -R :ops /srv/samba/ops
lens@lens-server:~$ sudo chmod 2770 /srv/samba/ops
lens@lens-server:~$ sudo chmod 1777 /srv/samba/shared
lens@lens-server:~$ sudo chown root:dev /srv/samba/collab-dev-sec
[sudo] password for lens:
Sorry, try again.
[sudo] password for lens:
lens@lens-server:~$ sudo chmod 770 /srv/samba/collab-dev-sec
lens@lens-server:~$ sudo setfacl -m g:sec:r-- /srv/samba/collab-dev-sec
setfacl: /srv/samba/collab-dev-sec: Operation not permitted
lens@lens-server:~$ sudo setfacl -m g:sec:r-- /srv/samba/collab-dev-sec
lens@lens-server:~$
```

5. Tests fonctionnels

```
lens@lens-server:~$ id Tom
uid=1001(Tom) gid=1004(Tom) groups=1004(Tom),1001(dev)
lens@lens-server:~$ cd /srv/samba/ops
bash: cd: /srv/samba/ops: Permission denied
lens@lens-server:~$ cd /srv/samba/sec
bash: cd: /srv/samba/sec: Permission denied
```



6. Audit complet des permissions

```
Dec 6 20:10
lens@lens-server: ~
lens@lens-server:~$ sudo ls -lR /srv/company
[sudo] password for lens:
ls: cannot access '/srv/company': No such file or directory
lens@lens-server:~$ sudo ls -lR /srv/samba
/srv/samba:
total 24
drwxrwx---+ 2 root dev 4096 Dec 6 18:07 collab-dev-sec
drwxrws--- 2 root dev 4096 Dec 6 18:06 dev
drwxrws--- 2 root ops 4096 Dec 6 18:06 ops
drwxrwxrwx 2 root root 4096 Dec 6 18:07 public
drwxrws--- 2 root sec 4096 Dec 6 20:01 sec
drwxrwxrwt 2 root root 4096 Dec 6 18:07 shared

/srv/samba/collab-dev-sec:
total 0

/srv/samba/dev:
total 0

/srv/samba/ops:
total 0

/srv/samba/public:
total 0

/srv/samba/sec:
total 4
-rw-rw-r-- 1 ralph sec 25 Dec 6 20:01 lp.txt

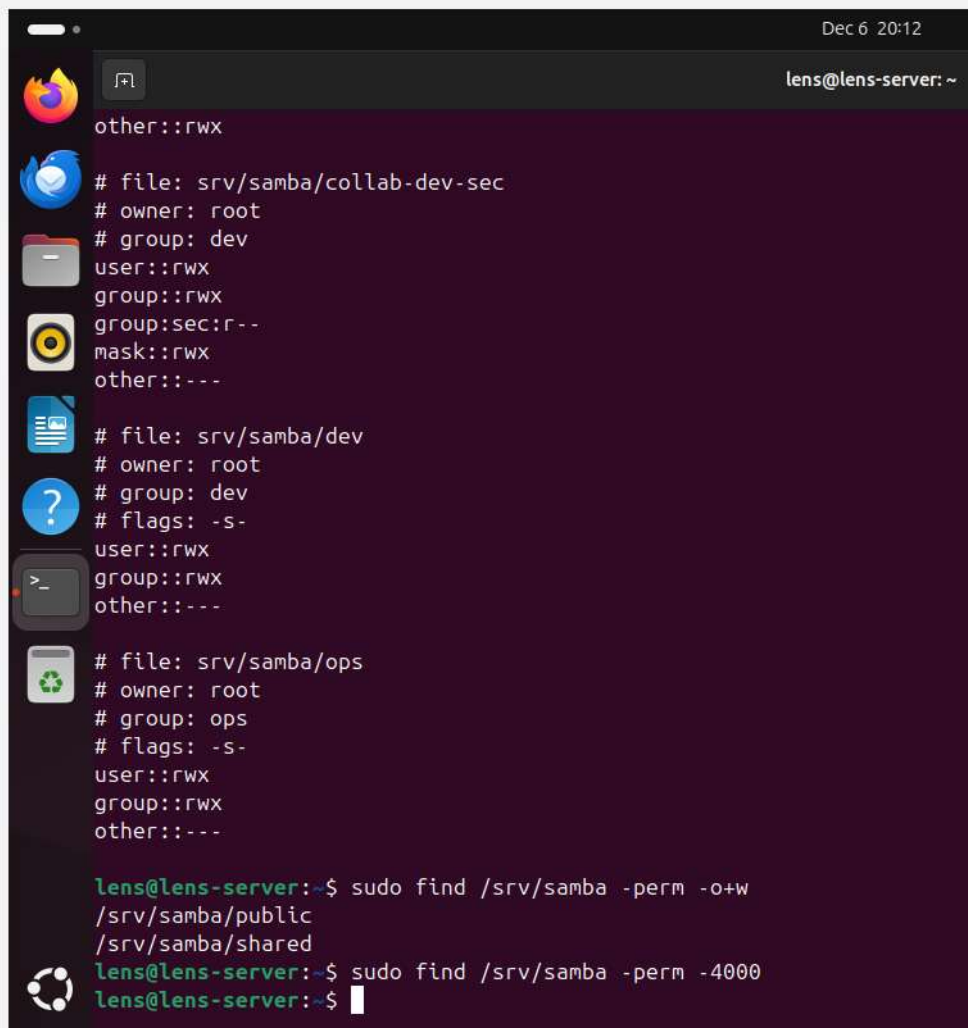
/srv/samba/shared:
total 0
lens@lens-server:~$ sudo getfacl -R /srv/samba
getfacl: Removing leading '/' from absolute path names
```

```
Dec 6 20:11
lens@lens-server: ~
/srv/samba/shared:
total 0
lens@lens-server:~$ sudo getfacl -R /srv/samba
getfacl: Removing leading '/' from absolute path names
# file: srv/samba
# owner: root
# group: root
user::rwx
group::r-x
other::r-x

# file: srv/samba/public
# owner: root
# group: root
user::rwx
group::rwx
other::rwx

# file: srv/samba/sec
# owner: root
# group: sec
# flags: -s-
user::rwx
group::rwx
other::---

# file: srv/samba/sec/lp.txt
# owner: ralph
# group: sec
user::rw-
group::rw-
other::r--
```

```
lens@lens-server: ~  
other::rwx  
# file: /srv/samba/collab-dev-sec  
# owner: root  
# group: dev  
user::rwx  
group::rwx  
group:sec:r--  
mask::rwx  
other::---  
# file: /srv/samba/dev  
# owner: root  
# group: dev  
# flags: -s-  
user::rwx  
group::rwx  
other::---  
# file: /srv/samba/ops  
# owner: root  
# group: ops  
# flags: -s-  
user::rwx  
group::rwx  
other::---  
lens@lens-server:~$ sudo find /srv/samba -perm -o+w  
/srv/samba/public  
/srv/samba/shared  
lens@lens-server:~$ sudo find /srv/samba -perm -4000  
lens@lens-server:~$
```

Cocclusion.

Ce travail m'a permis comprendre les comptes utilisateurs et groupes sous Linux, créer, modifier, supprimer utilisateurs et groupes, gérer les permissions : r, w, x, chmod, chown, chgrp, configurer les permissions avancées : SUID, SGID, sticky bit, savoir vérifier et auditer les droits. Donc la tâche est réussie.

